

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

53002007 - Energy Resources Fundamentals

DEGREE PROGRAMME

05BK - Máster Universitario En Ingeniería De La Energía

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	53002007 - Energy Resources Fundamentals
No of credits	3 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	05BK - Máster Universitario en Ingeniería de la Energía
Centre	05 - E.T.S. De Ingenieros Industriales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
David Leon Ruiz		david.leon.ruiz@upm.es	M - 09:00 - 12:00 Tu - 09:00 - 12:00 Tutorial times are approximate and may be subject to change at any time. Tutoring times should be confirmed by emailing the professor.

Ljiljana Medic Pejic (Subject coordinator)	417	liliana.medic@upm.es	Tu - 11:00 - 14:00 W - 11:00 - 14:00 Tutorial times are approximate and may be subject to change at any time. Tutoring times should be confirmed by emailing the professor.
Javier Garcia Torrent	419	javier.garcia@upm.es	Tu - 09:30 - 13:30 Th - 17:30 - 19:30 Tutorial times are approximate and may be subject to change at any time. Tutoring times should be confirmed by emailing the professor.
Blanca Castells Somoza	422	b.castells@upm.es	M - 15:00 - 17:00 F - 08:00 - 12:00 Tutorial times are approximate and may be subject to change at any time. Tutoring times should be confirmed by emailing the professor.

Isabel Amez Arenillas	427	isabel.amez@upm.es	M - 08:00 - 10:00 Tu - 08:00 - 10:00 W - 08:00 - 10:00 Tutorial times are approximate and may be subject to change at any time. Tutoring times should be confirmed by emailing the professor.
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* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Fluid mechanics
- Organic and Inorganic Chemistry
- Differential equations
- Thermodynamics
- Heat and mass transfer

4. Skills and learning outcomes *

4.1. Skills to be learned

CB10 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CB8 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios

CB9 - Que los estudiantes sepan comunicar sus conclusiones y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades

CE12 - Aplicar conocimientos y disponer de habilidades para acometer el diseño, control y análisis de procesos industriales basados en la generación de calor por combustión convencional y avanzada, evaluando los combustibles mejor adaptados a cada aplicación y proponer soluciones razonadas en el empleo de combustibles

CE6 - Disponer de habilidades, criterios y conocimientos para analizar de forma objetiva el impacto ambiental de cualquier fuente de energía.

CE9 - Disponer de criterios y herramientas para entender la composición y características de los diferentes tipos de combustibles convencionales y no convencionales.

CG1 - Aplicar conocimientos de ciencias y tecnologías avanzadas a la práctica profesional o investigadora de la Ingeniería Energética.

CG5 - Comprender el impacto de la Ingeniería Energética en el medio ambiente, el desarrollo sostenible de la sociedad y la importancia de trabajar en un entorno profesional y responsable.

4.2. Learning outcomes

RA136 - Conocer la situación actual, evolución hasta el momento y perspectivas futuras de los diferentes mercados de energía

RA16 - Comprender los problemas medioambientales de la producción energética

RA236 - Evaluar los efectos económicos y ambientales de la implantación de escenarios energéticos

RA17 - Análisis y resolución de problemas de sostenibilidad en el entorno de las energías renovables

RA275 - RA213 - RA101 - Comprender los mecanismos y procesos físico-químicos y geológicos así como su aplicación a la hora de determinar el comportamiento de los hidrocarburos en el medio natural.

RA105 - Explicar y evaluar las características, fortalezas y debilidades, y alternativas de vectores energéticos disponibles en un contexto de descarbonización del sistema energético

RA26 - Habilidad de cálculos relativos a procesos de combustión y sus combustibles

RA113 - Relacionar el funcionamiento en el motor de los biocombustibles con sus ventajas e inconvenientes medioambientales, respecto de los combustibles tradicionales

RA114 - Formular opiniones, a partir de información incompleta o limitada, sobre las responsabilidades sociales y éticas vinculadas a los cultivos energéticos

RA112 - Aplicar los procesos de producción de biocombustibles

RA267 - Redactar, organizar y presentar trabajos científicos.

RA111 - Conocer las normas de calidad de los biocombustibles

RA110 - Analizar las implicaciones sociales y ambientales de los biocombustibles

RA101 - Identificar, plantear y resolver problemas de optimización en ingeniería.

RA281 - RA26 - Habilidad de cálculos relativos a procesos de combustión y sus combustibles

RA198 - Aplicar las tecnologías de captura y almacenamiento de CO₂

RA200 - Analizar las aplicaciones de la combustión en la generación de energía

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Resultados de Aprendizaje

RA26 - Habilidad de cálculos relativos a procesos de combustión y sus combustibles

RA105 - Explicar y evaluar las características, fortalezas y debilidades, y alternativas de vectores energéticos disponibles en un contexto de descarbonización del sistema energético

RA110 - Analizar las implicaciones sociales y ambientales de los biocombustibles

RA111 - Conocer las normas de calidad de los biocombustibles

RA112 - Aplicar los procesos de producción de biocombustibles

RA113 - Relacionar el funcionamiento en el motor de los biocombustibles con sus ventajas e inconvenientes medioambientales, respecto de los combustibles tradicionales

RA114 - Formular opiniones, a partir de información incompleta o limitada, sobre las responsabilidades sociales y éticas vinculadas a los cultivos energéticos

RA213 - RA101 - Comprender los mecanismos y procesos físico-químicos y geológicos así como su aplicación a la hora de determinar el comportamiento de los hidrocarburos en el medio natural.

RA276 - RA267 - Redactar, organizar y presentar trabajos científicos.

RA278 - RA15 - Analizar el concepto de sostenibilidad ambiental, económica y de recursos

5.2. Syllabus

1. The origin and composition of the fuels
2. Solid fuels: Coal and Biomass
3. Petroleum and renewable liquid fuels
4. Natural gas and gaseous fuels from renewable energy sources
5. Thermochemistry of combustion
6. Propagation of Reaction Front: Detonation and Deflagration
7. Theory of the flame
 - 7.1. Laminar and turbulent pre-mixed flames
 - 7.2. Diffusion flames, laminar and turbulent
8. Models and mechanisms of the combustion process
9. The formation of emissions in the combustion process
10. Global & European energy transition outlooks, projections and scenarios

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	The origin and composition of the fuels Duration: 02:00 Lecture			Evaluación continua sumativa Other assessment Progressive assessment Not Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05
2	Solid fuels: Coal and Biomass Duration: 02:00 Lecture			Evaluación continua sumativa Other assessment Progressive assessment Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05
3	Solid fuels: Coal and Biomass Duration: 02:00 Lecture			Evaluación continua sumativa Other assessment Progressive assessment Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05
4	Petroleum and renewable liquid fuels Duration: 02:00 Lecture			Evaluación continua sumativa Other assessment Progressive assessment Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05

5	Petroleum and renewable liquid fuels Duration: 02:00 Lecture			Evaluación continua sumativa Other assessment Progressive assessment Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05
6	Natural gas and gaseous fuels from renewable energy sources Duration: 02:00 Lecture			Autoevaluación Plataforma Moodle Online test Progressive assessment Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05
7	Natural gas and gaseous fuels from renewable energy sources Duration: 02:00 Lecture			Autoevaluación Plataforma Moodle Online test Progressive assessment Not Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05
8	Thermochemistry of combustion Duration: 02:00 Lecture			Autoevaluación Plataforma Moodle Online test Progressive assessment Not Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05
9	Thermochemistry of combustion Duration: 02:00 Lecture			Autoevaluación Plataforma Moodle Online test Progressive assessment Not Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05

10	Propagation of Reaction Front: Detonation and Deflagration Duration: 02:00 Lecture			Autoevaluación Plataforma Moodle Online test Progressive assessment Not Presential Duration: 00:30 Pruebas de repaso en el aula Written test Progressive assessment Presential Duration: 00:05
11	Theory of the flame Duration: 01:00 Lecture Theory of the flame Duration: 01:00 Problem-solving class			Trabajos tutelados Individual presentation Progressive assessment Presential Duration: 00:30
12	Laminar and turbulent pre-mixed flames Duration: 01:00 Lecture Diffusion flames, laminar and turbulent Duration: 01:00 Problem-solving class			Trabajos tutelados Individual presentation Progressive assessment Presential Duration: 00:30
13	Models and mechanisms of the combustion process Duration: 01:00 Lecture The formation of emissions in the combustion process Duration: 01:00 Problem-solving class			Trabajos tutelados Individual presentation Progressive assessment Presential Duration: 00:30
14	Global & European energy transition outlooks, projections and scenarios Duration: 01:00 Lecture Global & European energy transition outlooks, projections and scenarios Duration: 01:00 Problem-solving class			Trabajos tutelados Individual presentation Progressive assessment Presential Duration: 00:30
15				
16				
17				Evaluación final Written test Progressive assessment Presential Duration: 02:00 Examen final Written test Global examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	Evaluación continua sumativa	Other assessment	No Presential	00:30	.5%	/ 10	CE6 CG1 CG5 CB10 CE9
1	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CB10 CE9
2	Evaluación continua sumativa	Other assessment	Face-to-face	00:30	.5%	/ 10	CE6 CG1 CG5 CB10 CE9 CE12
2	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CB10 CE9
3	Evaluación continua sumativa	Other assessment	Face-to-face	00:30	.5%	/ 10	CG5 CB10 CE9 CE12 CE6 CG1
3	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CE9 CB10
4	Evaluación continua sumativa	Other assessment	Face-to-face	00:30	.5%	/ 10	CE6 CG1 CG5 CE12 CB8
4	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CB10 CE9
5	Evaluación continua sumativa	Other assessment	Face-to-face	00:30	.5%	/ 10	CE6 CG1 CG5 CE12

5	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CB10 CE9
6	Autoevaluación Plataforma Moodle	Online test	Face-to-face	00:30	.5%	/ 10	CE6 CG1 CG5 CE12
6	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CB10 CE9
7	Autoevaluación Plataforma Moodle	Online test	No Presential	00:30	.5%	/ 10	CE6 CG1 CG5 CE12
7	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CB10 CE9
8	Autoevaluación Plataforma Moodle	Online test	No Presential	00:30	.5%	/ 10	CE6 CG1 CG5 CE12
8	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CB10 CE9
9	Autoevaluación Plataforma Moodle	Online test	No Presential	00:30	.5%	/ 10	CG1 CG5 CE12 CE6
9	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CB10 CE9
10	Autoevaluación Plataforma Moodle	Online test	No Presential	00:30	.5%	/ 10	CG5 CE12 CE6 CG1
10	Pruebas de repaso en el aula	Written test	Face-to-face	00:05	.5%	/ 10	CB10 CE9
11	Trabajos tutelados	Individual presentation	Face-to-face	00:30	5%	/ 10	CE12 CB8 CB9
12	Trabajos tutelados	Individual presentation	Face-to-face	00:30	5%	/ 10	CE12 CB8 CB9
13	Trabajos tutelados	Individual presentation	Face-to-face	00:30	5%	/ 10	CE12 CB8 CB9
14	Trabajos tutelados	Individual presentation	Face-to-face	00:30	5%	/ 10	CB9 CE12 CB8
17	Evaluación final	Written test	Face-to-face	02:00	70%	3.5 / 10	CE6 CG1 CG5 CB10 CE9 CE12 CB8

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7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Examen final	Written test	Face-to-face	02:00	100%	3.5 / 10	CE6 CG1 CG5 CB10 CE9 CE12 CB8 CB9

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Examen Convocatoria extraordinaria	Written test	Face-to-face	03:00	100%	/ 10	CG5 CB10 CE9 CE12 CB8 CB9 CE6 CG1

7.2. Assessment criteria

In order to pass the course it is essential to

- Score at least 3.5 points (out of 10) in the final exam.

The exam may consist of the following exercises:

1) Short questions, which require the application of the theory taught. These will include, among others, closed-answer, multiple-choice questions. These are simple applications of fundamental concepts from the subject syllabus. In

In some cases they involve some simple calculation by applying an equation, or defining when an equation is applicable or what the meaning of the variables involved is.

2) Application problems or exercises, with the same characteristics as those proposed and solved during the course.

Assessment by final examination only

Students who do not wish to take the partial tests, and therefore opt for assessment by "final test only", must additionally take a final exam on all the achievement indicators. This final exam will have the same characteristics as the continuous assessment modality.

Extraordinary call

All students, both continuous assessment and final exam students, who have not passed the ordinary exam may sit the extraordinary exam, in which they will be examined for the whole course. The exam will have the same characteristics as the "final exam only".

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Forman A. Williams	Bibliography	Combustion Theory
Thierry Poinot	Bibliography	Theoretical Numerical Combustion
J. Warnatz	Bibliography	Physical and Chemical Fundamentals, Modeling and Simulation
GLASSMAN, I., YETTER, R.	Bibliography	Combustion. Academic Press Elsevier. 2008
LOWRY, H.H.	Bibliography	Chemistry of Coal Utilization. John Wiley & Son, Londres
GARDINER, W. C. Jr.	Bibliography	Combustion Chemistry. Springer, 1984.
Plataforma Moodle: asignatura Fundamentos de Recursos Energéticos. En la misma se hacen referencia y vínculos a otros recursos web	Web resource	
J. N. Bradley	Bibliography	Flame and Combustion Phenomena

9. Other information

9.1. Other information about the subject

The course is designed for continuous assessment, with numerous classroom activities to apply the content of the programme. Throughout the course there will be collaborative work, exercises, Moodle questionnaires, application and assessment tests. Due to the technical nature of the subject, it is not recommended that students be assessed solely by a final exam.